

Deep Learning

Backpropagation

Recap

- Gradient of a scalar valued function $f(\mathbf{x})$: $\mathbf{x} \rightarrow \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_D} \right)$

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- Gradient of a vector valued function $\mathbf{f}(\mathbf{x})$ is called Jacobian:

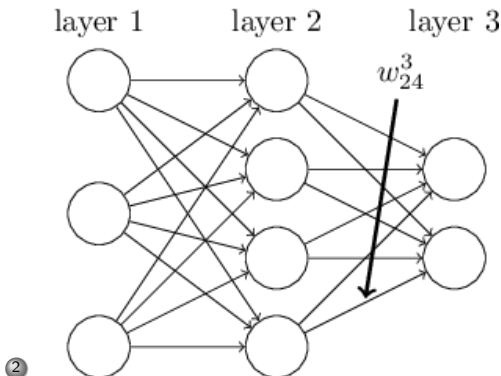
$$\mathbf{J} = \begin{bmatrix} \frac{\partial \mathbf{f}}{\partial x_1} & \dots & \frac{\partial \mathbf{f}}{\partial x_n} \end{bmatrix} = \begin{bmatrix} \nabla^T f_1 \\ \vdots \\ \nabla^T f_m \end{bmatrix} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \dots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}$$

MLP: Some Notation

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- ④ Vector of activations (or, biases) at a layer l is denoted by a bold-faced $\mathbf{x}^l$ (or $\mathbf{b}^l$) and W^l is the matrix of weights into layer l

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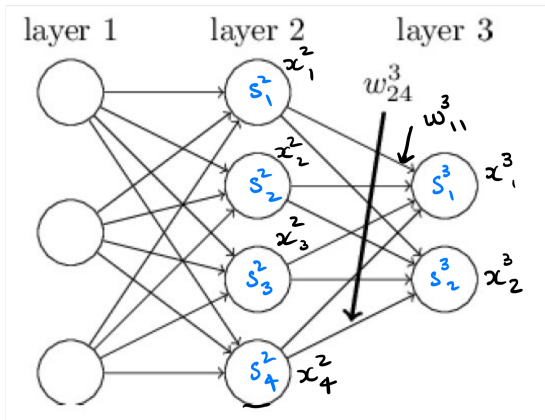
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- ④ σ is the activation function that applies element-wise

Neural network nomenclature



Gradient descent on MLP

- Loss is $\mathcal{L}(W, \mathbf{b}) = \sum_n l(f(x_n; W, \mathbf{b}), y_n) = \sum_n l(\mathbf{x}^L, y_n)$ (L is the number of layers in the MLP)

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- For applying Gradient descent, we need gradient of individual sample loss with respect to all the model parameters

$$l_n = l(f(x_n; W, \mathbf{b}), y_n)$$

$$\frac{\partial l_n}{\partial W_{jk}^{(l)}} \text{ and } \frac{\partial l_n}{\partial \mathbf{b}_j^{(l)}} \text{ for all layers } l$$

Forward pass operation

$$x^{(0)} = x \xrightarrow{W^{(1)}, \mathbf{b}^{(1)}} s^{(1)} \xrightarrow{\sigma} x^{(1)} \xrightarrow{W^{(2)}, \mathbf{b}^{(2)}} s^{(2)} \dots x^{(L-1)} \xrightarrow{W^{(L)}, \mathbf{b}^{(L)}} s^{(L)} \xrightarrow{\sigma} x^{(L)} = f(x; W, \mathbf{b})$$

Formally, $x^{(0)} = x, f(x; W, \mathbf{b}) = x^{(L)}$

$$\forall l = 1, \dots, L \quad \begin{cases} s^{(l)} &= W^{(l)} x^{(l-1)} + \mathbf{b}^{(l)} \\ x^{(l)} &= \sigma(s^{(l)}) \end{cases}$$

Chain rule of differential calculus

- Core concept of backpropagation

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$$(f \circ g)'(x) = f'(g(x)) \cdot g'(x)$$

Chain rule of differential calculus

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$$\frac{\partial}{\partial x} f(g(x)) = \left. \frac{\partial f(a)}{\partial a} \right|_{a=g(x)} \cdot \frac{\partial g(x)}{\partial x}$$

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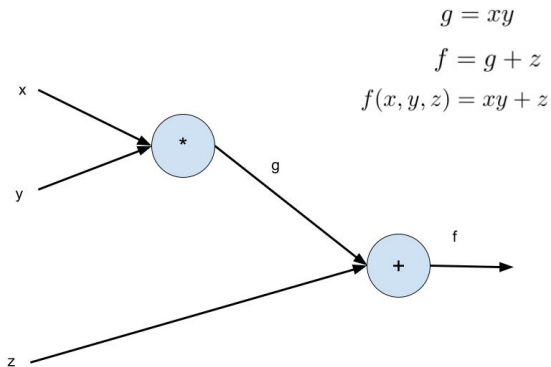
$$\textcircled{6} \quad \Delta y = \frac{\partial f}{\partial g_1(x)} \frac{dg_1(x)}{dx} \Delta x + \frac{\partial f}{\partial g_2(x)} \frac{dg_2(x)}{dx} \Delta x + \dots + \frac{\partial f}{\partial g_M(x)} \frac{dg_M(x)}{dx} \Delta x$$

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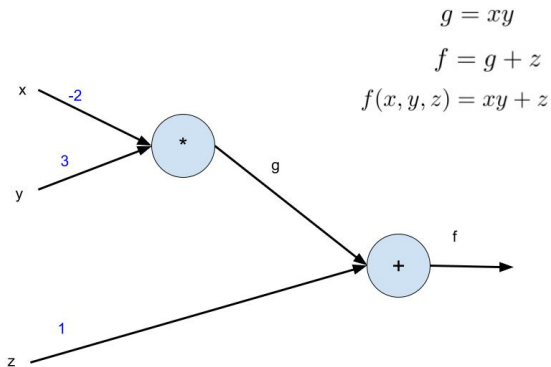
Chain rule of differential calculus

① $f(x) = e^{\sin(x^2)}$, let's find $\frac{\partial f}{\partial x}$

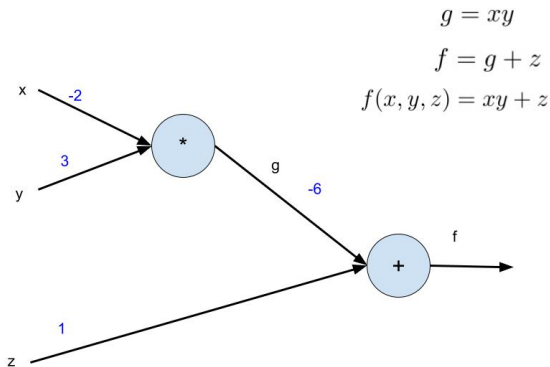
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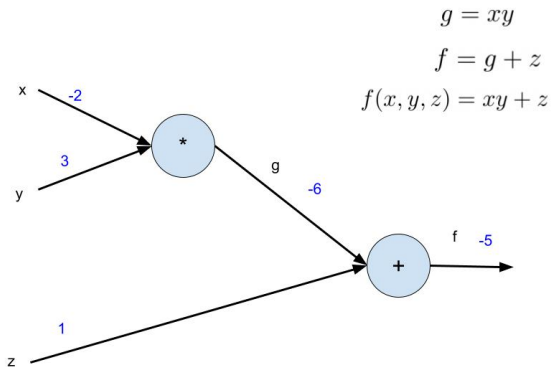
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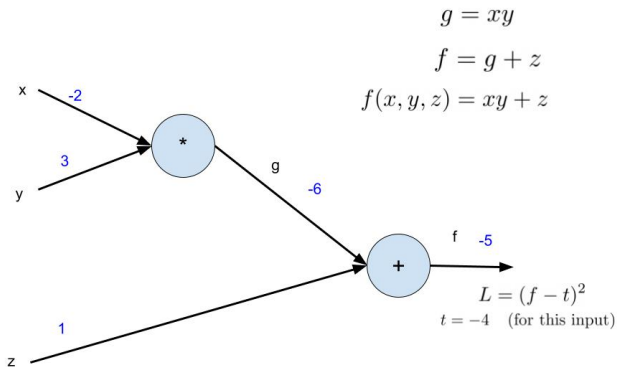
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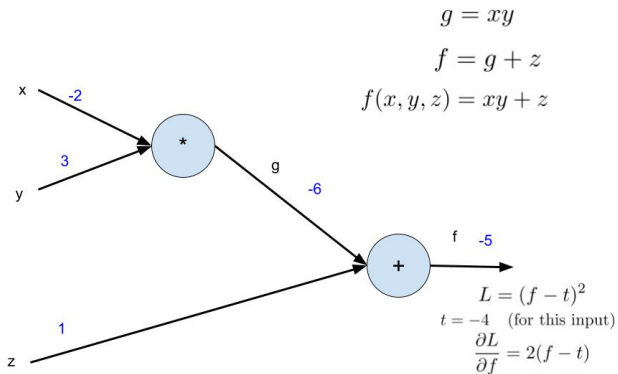
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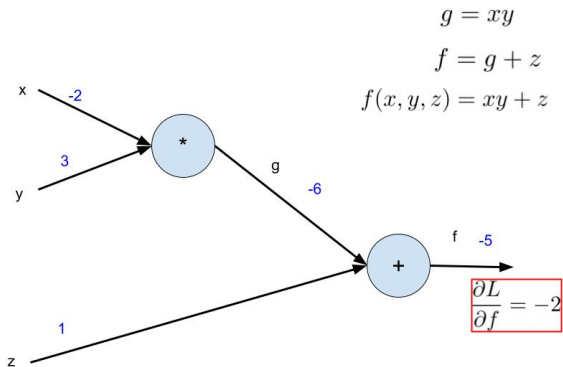
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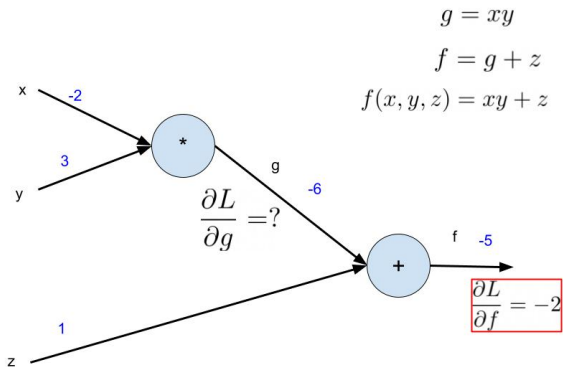
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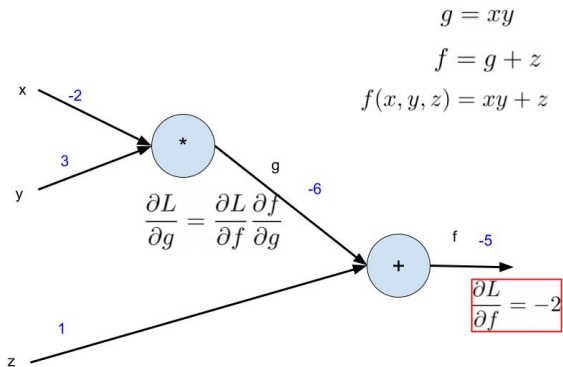
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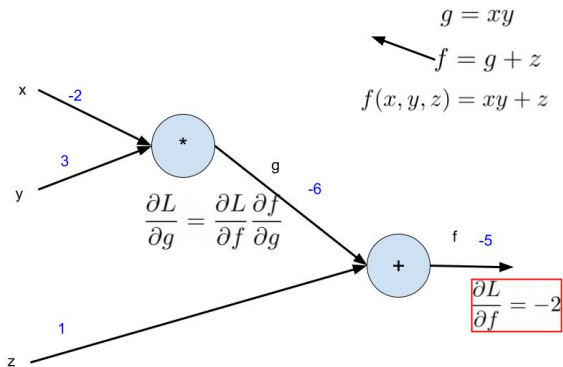
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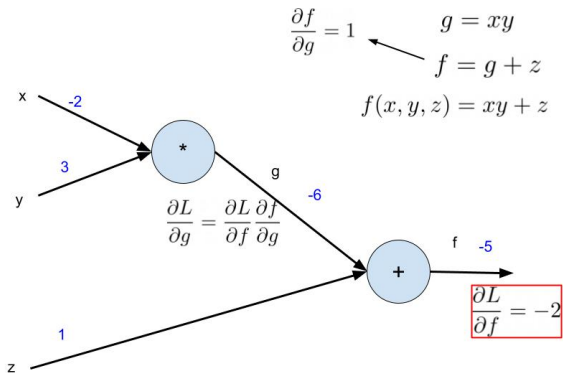
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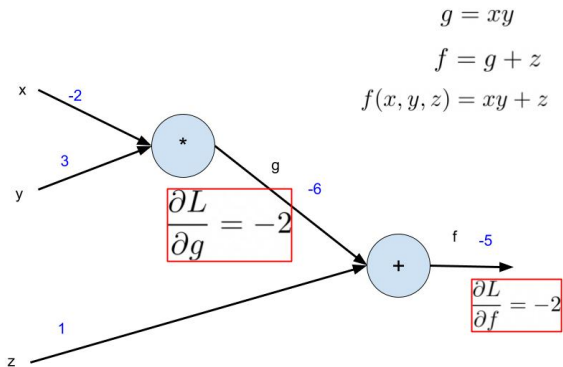
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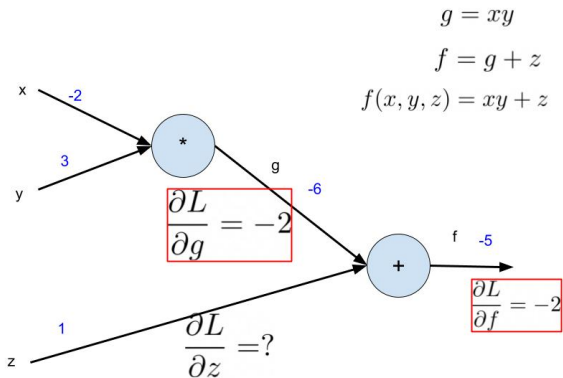
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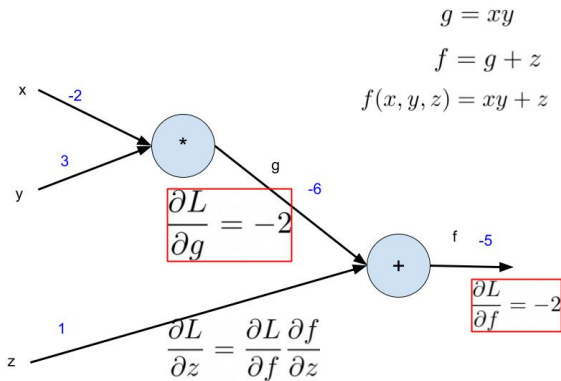
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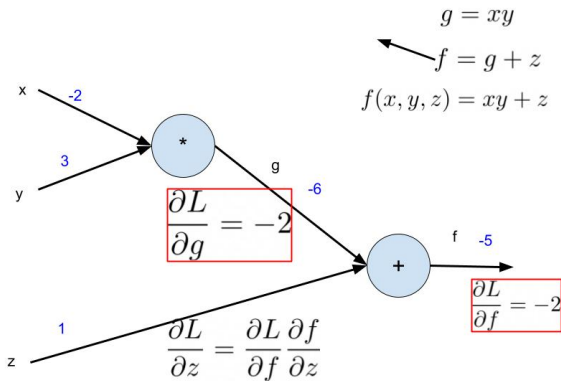
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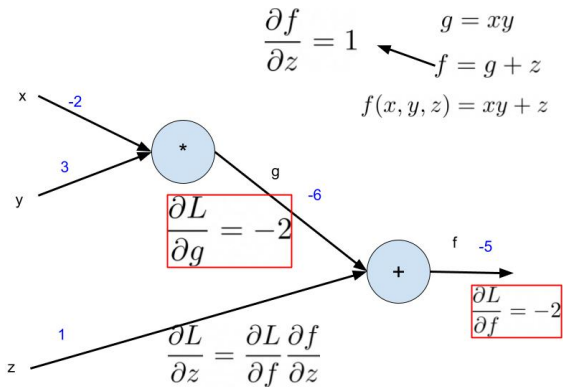
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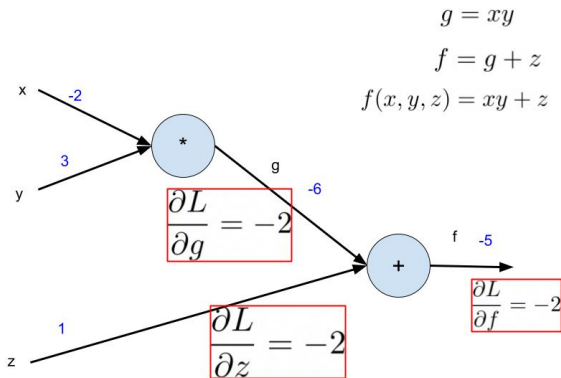
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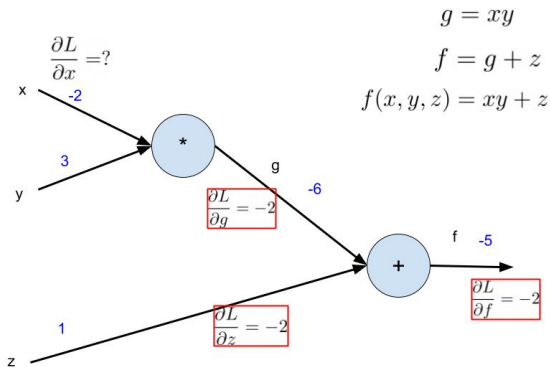
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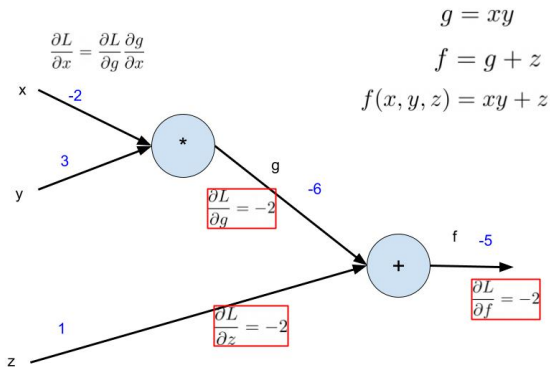
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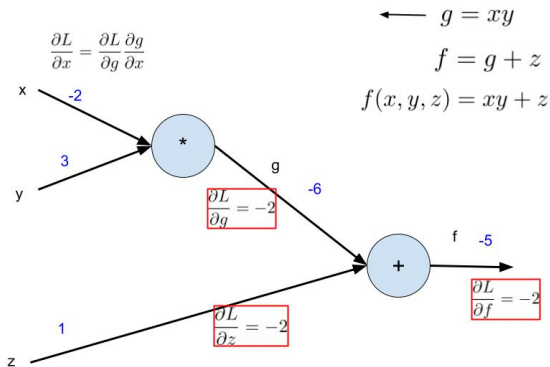
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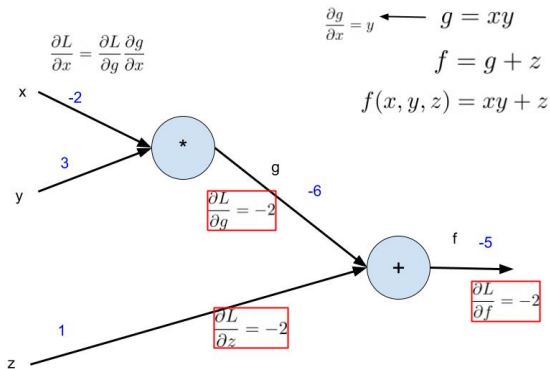
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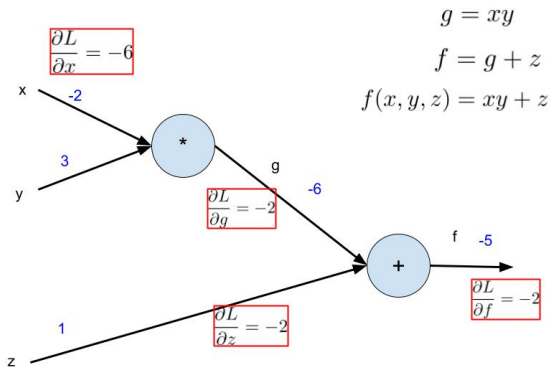
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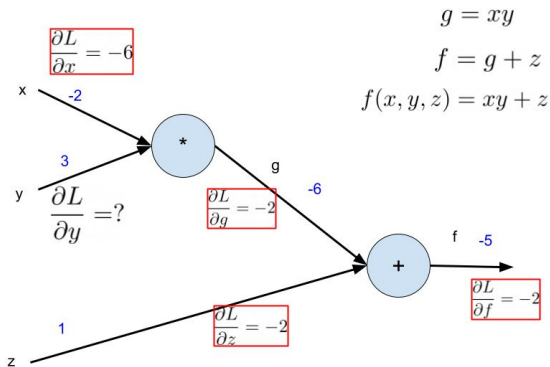
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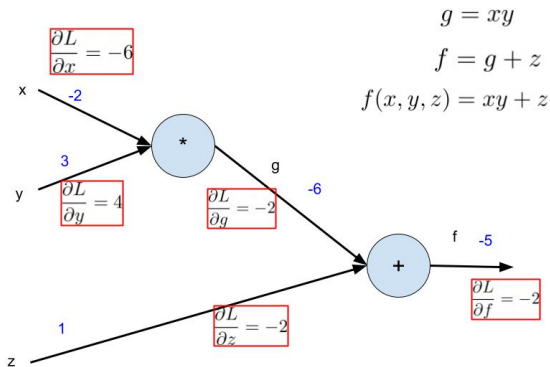
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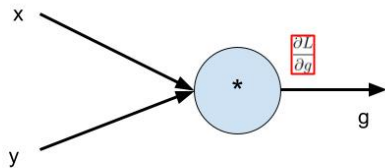
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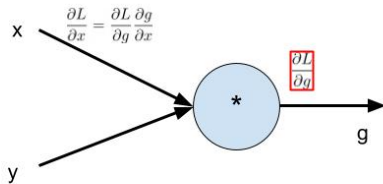
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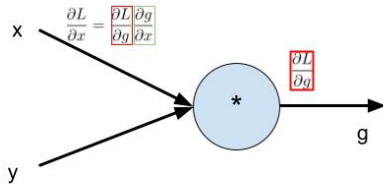
Gradient Flow



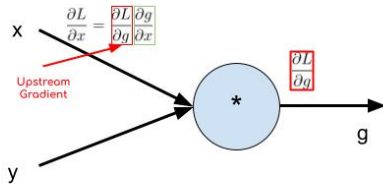
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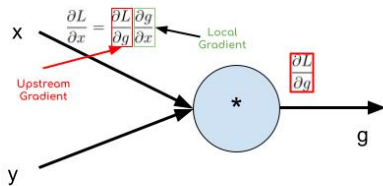
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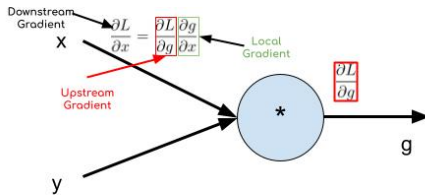
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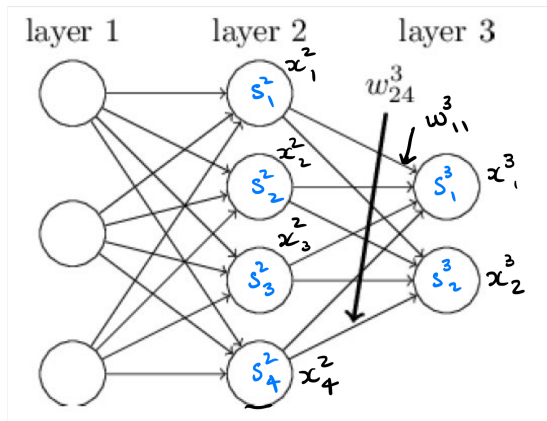
Chain rule of differential calculus for an MLP



$$J_{f_N \circ f_{N-1} \circ \dots \circ f_1(x)} = J_{f_N(f_{N-1}(\dots f_1(x)))} \cdot J_{f_{N-1}(f_{N-2}(\dots f_1(x)))} \cdot \dots \cdot J_{f_2(f_1(x))} \cdot J_{f_1(x)}$$

$J_{f(x)}$ is Jacobian of f computed at x .

Neural network nomenclature



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$$\frac{\partial \ell}{\partial s_i^{(l)}} = \frac{\partial \ell}{\partial x_i^{(l)}} \frac{\partial x_i^{(l)}}{\partial s_i^{(l)}} = \frac{\partial \ell}{\partial x_i^{(l)}} \sigma'(s_i^{(l)})$$

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We need gradients wrt parameters W and b

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- $$\frac{\partial \ell}{\partial b_i^{(l)}} = \frac{\partial \ell}{\partial s_i^{(l)}} \frac{\partial s_i^{(l)}}{\partial b_i^{(l)}} = \frac{\partial \ell}{\partial s_i^{(l)}} \quad (2)$$

Summary of Backprop

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$$\frac{\partial \ell}{\partial w_{i,j}^{(l)}} = \frac{\partial \ell}{\partial s_i^{(l)}} x_j^{(l-1)} \text{ and } \frac{\partial \ell}{\partial b_i^{(l)}} = \frac{\partial \ell}{\partial s_i^{(l)}}$$

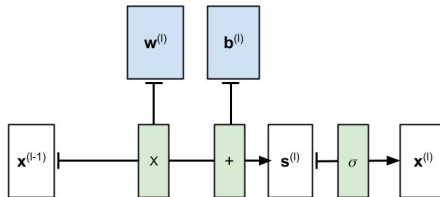
Jacobian in Tensorial form

$$\bullet \psi : \mathcal{R}^N \rightarrow \mathcal{R}^M \text{ then } \left[\frac{\partial \psi}{\partial x} \right] = \begin{bmatrix} \frac{\partial \psi_1}{\partial x_1} & \cdots & \frac{\partial \psi_1}{\partial x_N} \\ \vdots & \ddots & \vdots \\ \frac{\partial \psi_M}{\partial x_1} & \cdots & \frac{\partial \psi_M}{\partial x_N} \end{bmatrix}$$

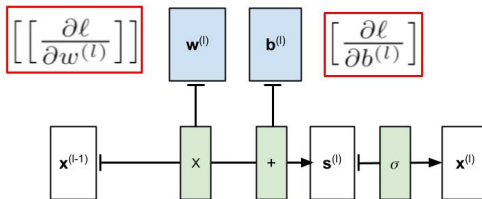
Jacobian in Tensorial form

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- $\psi : \mathcal{R}^{N \times M} \rightarrow \mathcal{R}$ then $\left[\left[\frac{\partial \psi}{\partial x} \right] \right] = \begin{bmatrix} \frac{\partial \psi}{\partial w_{1,1}} & \cdots & \frac{\partial \psi}{\partial w_{1,M}} \\ \vdots & \ddots & \vdots \\ \frac{\partial \psi}{\partial w_{N,1}} & \cdots & \frac{\partial \psi}{\partial w_{N,M}} \end{bmatrix}$

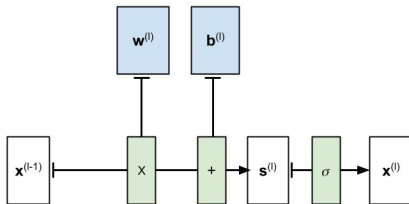
Forward Pass



Goal of Backward Pass

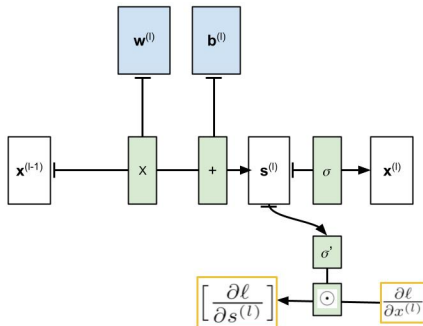


Begin from succeeding layer

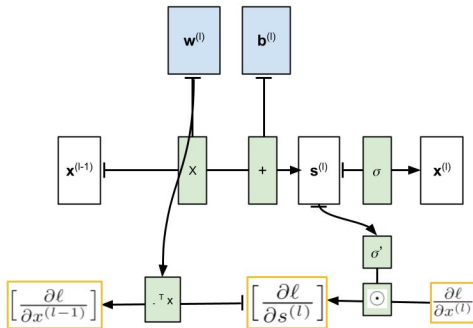


$$\boxed{\frac{\partial \ell}{\partial \mathbf{x}^{(l)}}}$$

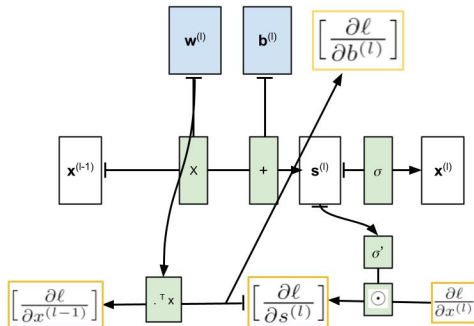
Begin from succeeding layer



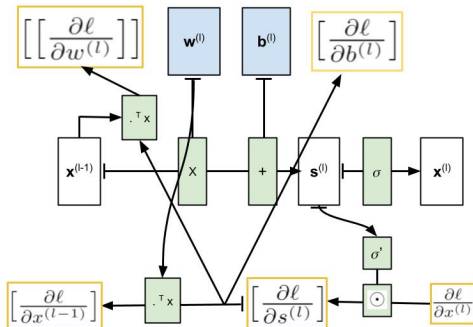
Begin from succeeding layer



Begin from succeeding layer



Begin from succeeding layer



Update the parameters

- $W^{(l)} = W^{(l)} - \eta \left[\left[\frac{\partial \ell}{\partial w^{(l)}} \right] \right]$ and $\mathbf{b}^{(l)} = \mathbf{b}^{(l)} - \eta \left[\frac{\partial \ell}{\partial b^{(l)}} \right]$

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- BP Needs all the intermediate layer results to be in memory
- Takes twice the computations of forward pass

High dimensional loss surfaces are complex

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- Can Gradient descent with BP find the 'right' function?
- Saddle points are far more frequent than local minima (exponential in network size)
- Most local minima are equivalent and lie close to the global minimum
- Active research!